Tushar Patil

PYTHON DEVELOPER

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PROFESSIONAL SUMMARY

- Python Backend Developer with 2.4+ years of experience in designing, developing, and deploying scalable RESTful APIs using Flask, Django REST Framework and FastAPI.
- Skilled in Python with expertise in **Flask, FastAPI** and **Django Rest Framework.**
- Experienced in managing SQL and NoSQL databases including PostgreSQL, MySQL, MongoDB, and legacy Oracle/SQL Server systems..
- Familiar with ReactJs**, HTML, CSS,** as well as libraries such as **NumPy** and Pandas.
- Well versed in tools like Postman for API testing and proficient in unit testing.
- Demonstrated experience in the development and automation of projects.
- Demonstrated expertise in leveraging **Agile Methodology** and efficiently managed projects.
- Utilized Git for version control, collaboration, and rollback support, facilitating seamless collaboration and ensuring release stability.

SKILL

- **Programming Languages:** Python, Bash
- **Frontend:** HTML5, CSS3, Bootstrap, JavaScript, Reactjs
- **Web Frameworks:** Django, Flask, FastAPI
- **API Development:** RESTful APIs, JSON, Postman
- **Database Technologies:** MySQL, SQLite, MongoDB, Postgress
- **ORM Tools:** SQLAlchemy
- **Data Handling & Analysis:** Pandas, NumPy, JSON, CSV, Excel integration
- **Version Control:** Git, GitHub
- **Security & Auth:** JWT, OAuth2, Permissions
- **Task Queues & Caching:** Redis, Celery
- **Development Practices:** OOP, TDD, MVC, PEP8 Compliance
- **Operating Platform:** Windows, Linux, Ubuntu

PROFESSIONAL EXPERIENCE

Organization: Markytics.AI, Pune Sep 2025 - Dec 2025

Role: Python Developer

- Contributed to backend development for fintech automation products involving **AI Voicebot** and **WhatsApp communication systems**.
- Developed the **CollectionOfficerTracker** feature using optimized PostgreSQL CTEs to generate branch-wise summaries (loan counts, officer counts, feedbacks, dialer logs) and detailed officer-level drilldowns.
- Designed and implemented **ETL pipelines** for multi-bank loan data using Pandas, including file-type auto-detection, column validation, data cleaning, merging, and standardized loading into PostgreSQL.
- Automated reporting workflows by generating Excel outputs and sending email summaries containing missing customer/loan lists and DPD/time-period categorizations.
- Implemented **bot instruction and rule logic** for AI VoiceBot flows, determining how the bot responds based on DPD, loan details, and call-stage conditions.
- Participated in configuration, validation, and debugging of Voicebot call flows, ensuring correct call triggers, feedback mapping, dialer count integration, and stable bot performance.
- Built the **summarization and intent-mapping** component that extracts repayment intent, callback requests, promise date & amount, and refusal reasons from conversation JSON.
- Collaborated with frontend, QA, and product teams using Git and Postman to support integration testing, documentation, and client onboarding processes.

Organization: BMS INFO SOLUTIONS Pvt. Ltd, Bangalore July 2023 - Sep 2025

Role: Software Developer

- Engineered scalable backend services using Flask, FastAPI and Django Rest FrameWork, reducing API response latency by 30%.
- Integrated multiple SQL/NoSQL data sources for seamless data processing and reporting.
- Automated ETL pipelines using Pandas and NumPy to transform raw sales data into structured datasets, cutting down processing time by 40%.
- Designed and implemented dashboards that improved visibility into KPIs, reducing manual reporting by 50%.
- Developed RESTful APIs, reducing frontend integration errors by 35% and improving data sync time.
- Administered and optimized PostgreSQL and MongoDB databases, optimizing queries for performance and scalability.
- Orchestrated multi-environment deployments using Docker on AWS, and Heroku.
- Performed rigorous debugging and API validation using Postman, with comprehensive unit tests written in Pytest and MagicMock to ensure application reliability and performance..
- Followed Agile workflows and collaborated via Git for streamlined code management.

PROJECT DETAILS

AI VoiceBot Communication Logic & Summarization Engine (Markytics Project)

- Developed backend logic for the AI VoiceBot system used by banks to automate customer communication and debt-collection workflows.
- Implemented **bot instruction and rule engine**, defining how the bot should respond based on customer profile, loan details, DPD, and call-stage logic.
- Built the **summarization and intent-mapping function**, extracting repayment intent, callback requests, promise-to-pay date & amount, and refusal reasons from conversation data.
- Mapped summarization output and customer intent into structured PostgreSQL fields, enabling real-time visibility on the SmartCollect dashboard for bank employees.
- Implemented **EOC (End of Call) logic**, ensuring calls terminate correctly based on timeouts, silence detection, or completion of bot rules.
- Integrated backend workflows to push VoiceBot results, customer insights, and conversation flags into frontend and reporting APIs.
- Debugged complete bot flows, validating rule execution, message mapping, and call outcomes across multiple banks and configurations.
- Collaborated with QA and operations teams to test customer scenarios, verify rule correctness, and resolve data inconsistencies.

Impact: Improved accuracy of customer intent detection (ready-to-pay, callback requests, refusal, wrong number), resulting in more reliable data for bank follow-up teams.
Tech Stack: Python, FastAPI/Django REST Framework, PostgreSQL, Redis, REST APIs, JSON Parsing, Git, Postman, Linux

Bank Data ETL & Automation Pipeline (Markytics Project)

- Built modular ETL pipelines for each bank, supporting file types such as **loan, customer, guarantor, overdue, and recovery**.
- Designed logic to **merge multi-file datasets** into a unified, structured dataset with correct relationships.
- implemented column validation, null checks, type correction, and rule-based transformations using Pandas.
- Added **duplicate detection**, optimized batch inserts, and DB mappings to existing customer/loan records.
- Integrated stored procedures for bank-specific rules and automated processing flows.
- implemented automated reporting: **generated Excel outputs and sent email summaries** highlighting missing customers/loans and categorized them by DPD.
- Added authentication, structured logging, error reporting, and secure REST APIs via Django REST Framework.

- Ensured the system gracefully handles bad input files, missing columns, format issues, and inconsistent naming patterns.

Impact: Improved file processing speed, reduced manual data cleanup efforts, and delivered a scalable ETL solution usable across different banks with minimal changes.
Tech Stack: Python, Django REST Framework, Pandas, PostgreSQL

Core Banking API System (BMS Project)
- Contributed to a modular core banking backend using Django REST Framework & PostgreSQL, owning the transaction & reconciliation module.
- Implemented double-entry ledger with safe repeatable transfers to avoid duplicate transactions
- Integrated **OAuth2.0 + MFA authentication**, **Vault-managed secrets**, and **AES-256 encryption** to meet banking-grade security standards.
- Reduced API response errors by **40%** and optimized ORM queries, improving fund transfer latency by **25%**.
- Achieved **>95% unit test coverage** with **Pytest & Django TestCase**, and improved collaboration by auto-generating **Swagger & Postman API collections**.

Impact: Enabled horizontal scaling with Celery workers and Dockerized services, handling **3x more transactions** without downtime.
Tech Stack: Python, Django REST Framework, PostgreSQL, Redis, Docker, AWS (EC2, S3), CI/CD, OAuth2.0, Vault

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EDUCATIONAL QUALIFICATION:

- **B.Tech(Computer):** D Y Patil School of engineering Ambi, D Y Patil University, Maharashtra
- **HSC:** D R Junior College, Nandurbar, Maharashtra
- **SSC:** D R High School, Nandurbar, Maharashtra

DECLARATION: The above information is true to the best of my knowledge.

Place: Pune
Date: / /2025

(Tushar Patil)